

Final Review

*Team Sternenjäger

TU Berlin Summer School 14/8/2026

Team introduction and responsibilities

Name	Responsibility	Launch day responsibility
Leila	-Maintain stable wireless communication -Collect and record real-time flight data -Process sensor data from CanSat	Program and debug all onboard and ground station code;ensure connection between modules and microcontrollers and successful operation of isolated cansat system; record real-time flight data.
Ana	-Design and test the whole electrical circuit -Check sensors, motors and power modules -Inspect battery status and power budget -Troubleshoot hardware electrical faults	Inspect all circuits, sensors and motors; check battery status; complete official launch operation Solder wires that came out
Lior	-Design the CanSat mechanical structure -Design and test the parachute system -Check mechanical fitting of electronic components	
Austen	Organize team tasks and assign work for each member -Assist with different system tests -Arrange tests and prepare for launch activities -Coordinate team communication within the group	Full pre-launch safety check; real-time flight status observation; track falling trajectory and landing point; monitor on-site safety; assist and lead the whole satellite recovery process.

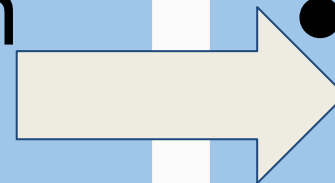


* Mission Recap

Mission Goals and Objective Outcomes

Goals:

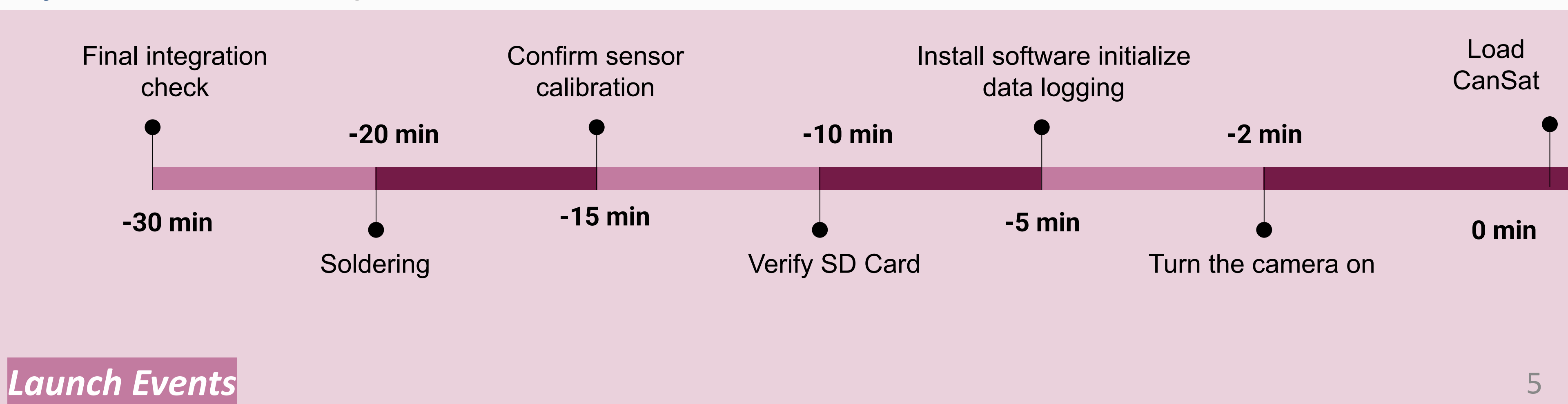
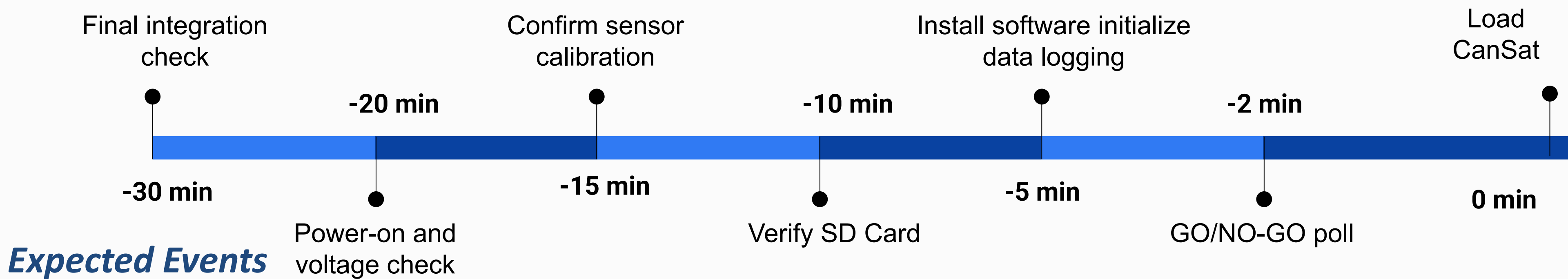
- Measure acceleration, altitude, and temperature in-flight
- Drive for 5 meters after touchdown
- Recover CanSat
- Data acquisition on pc
- Parachute detached from cansat

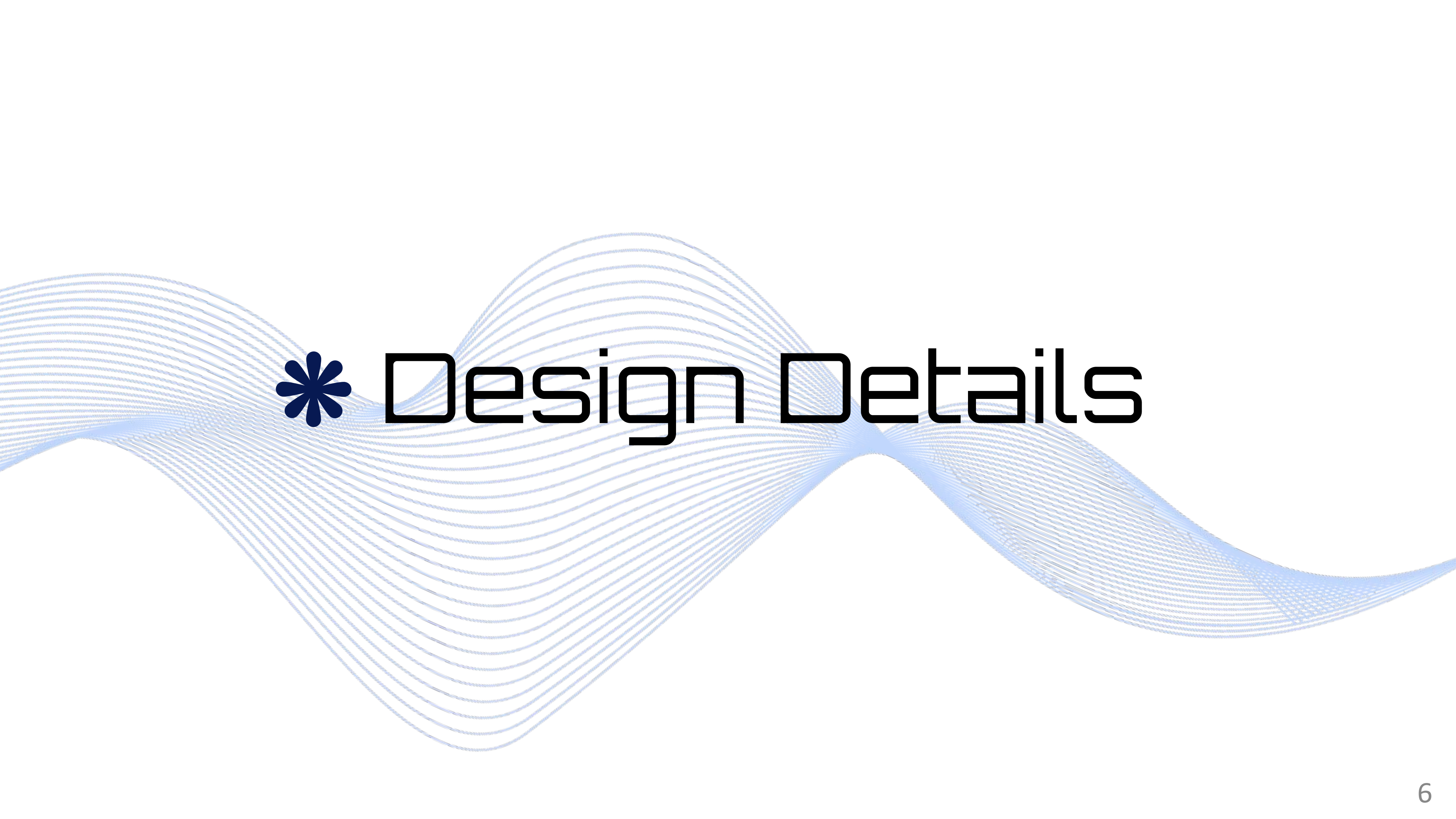


Outcomes:

- Green vs non-green objects detection and direction commands
- Data logging of Gyroscope, accelerometer, and temperature sensors
- CANSAT recovery

Flight Sequence Comparison

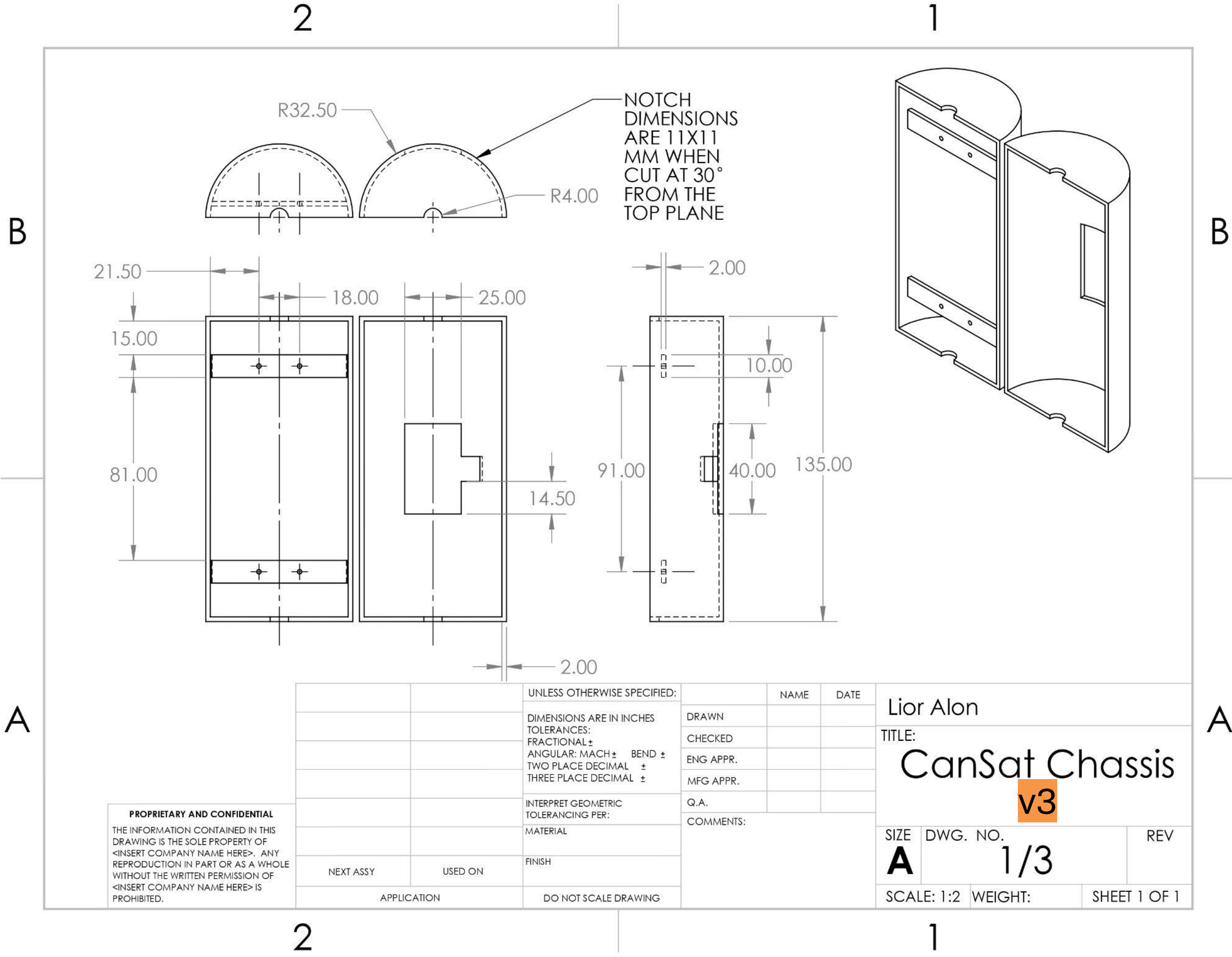




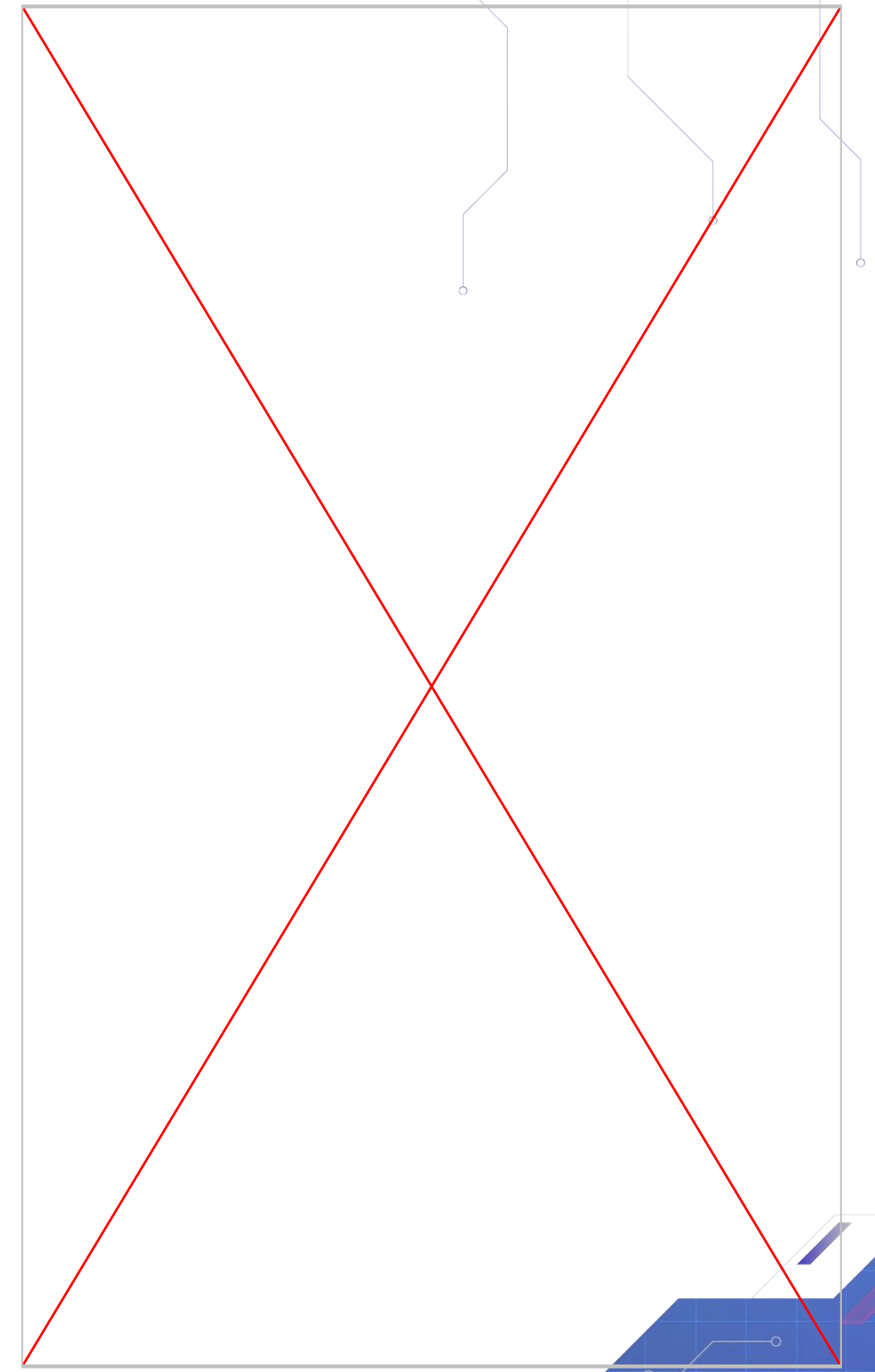
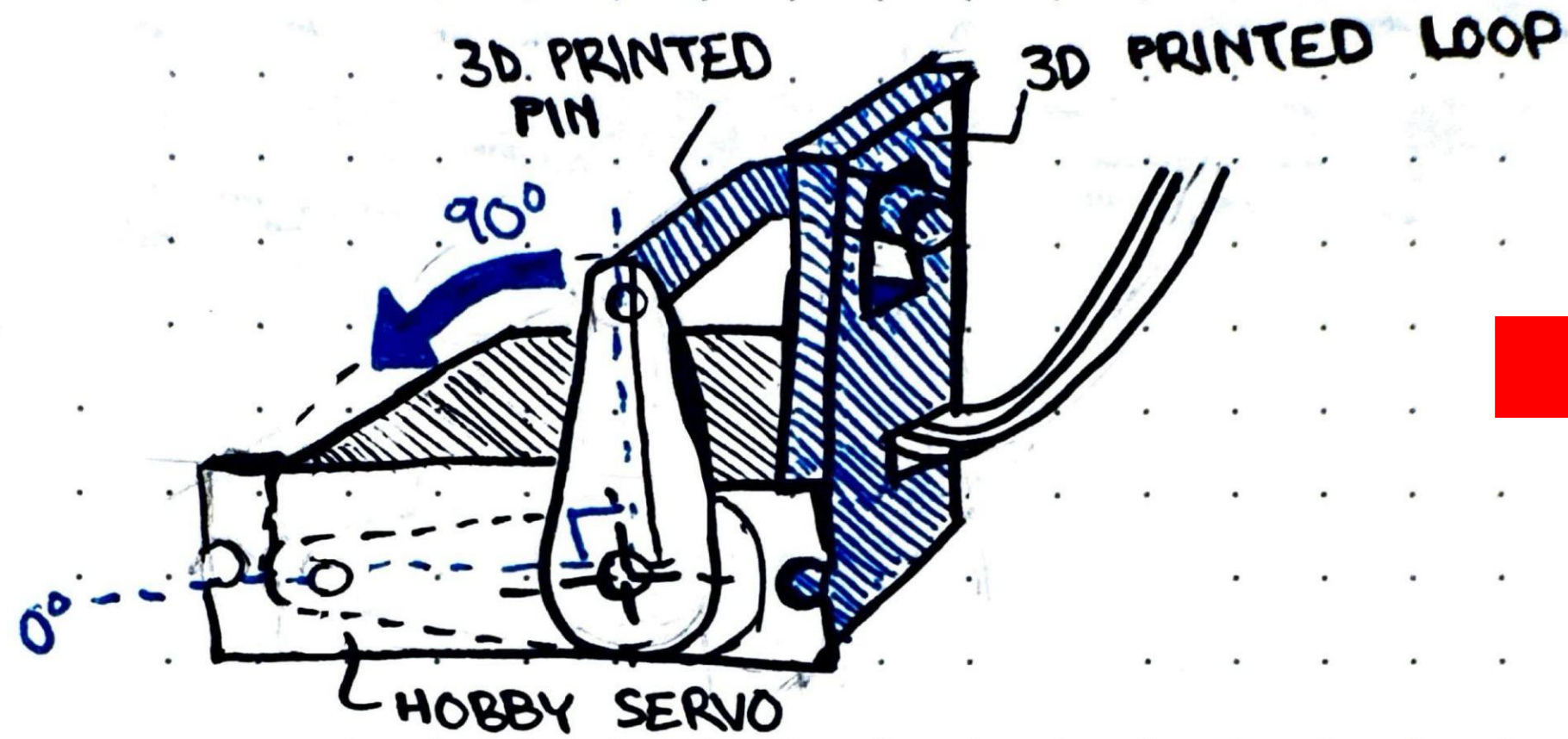
* Design Details

Final Prototype

- Added a notch in the chassis to secure the camera to the CanSat's exterior



Parachute Mechanism Demo

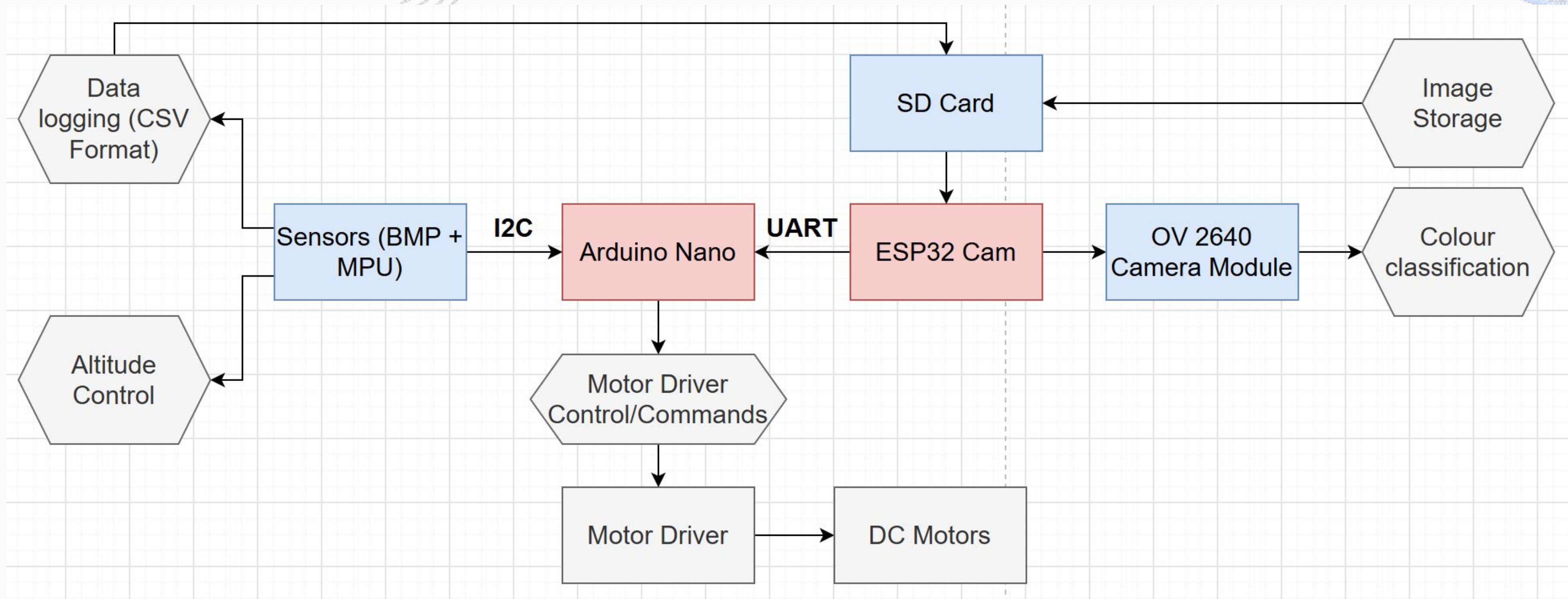


The background of the slide features abstract, flowing blue lines that create a sense of movement and depth. These lines are composed of many thin, parallel strokes that vary in density and direction, giving the impression of liquid or smoke. They are positioned around the central text, with some lines curving upwards from the bottom and others arching downwards from the top.

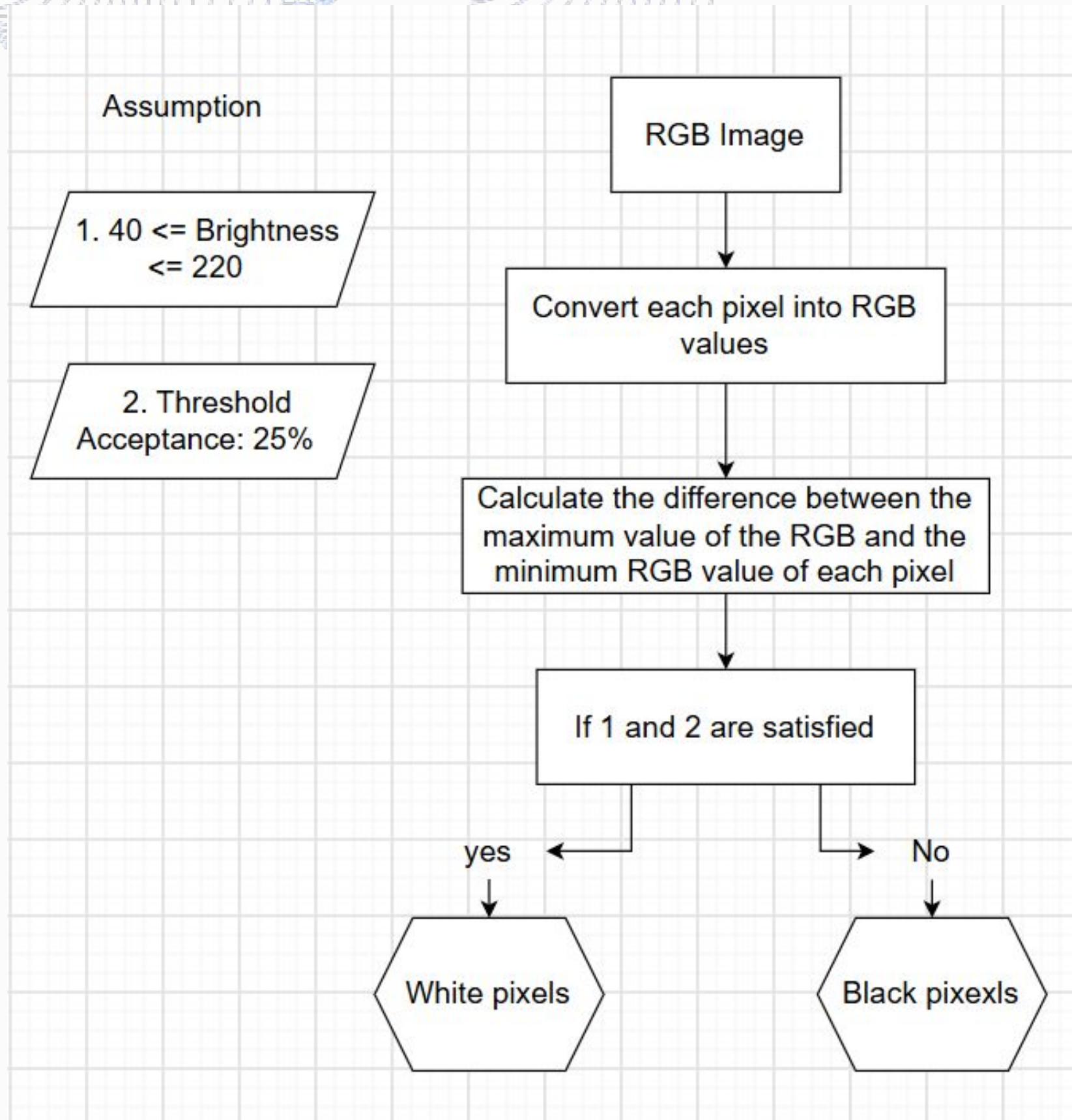
Flight Computer System Architecture & Testing

4-State Machine Architecture: Fully Autonomous Cansat Mission

05



Camera Colour Classification Logic



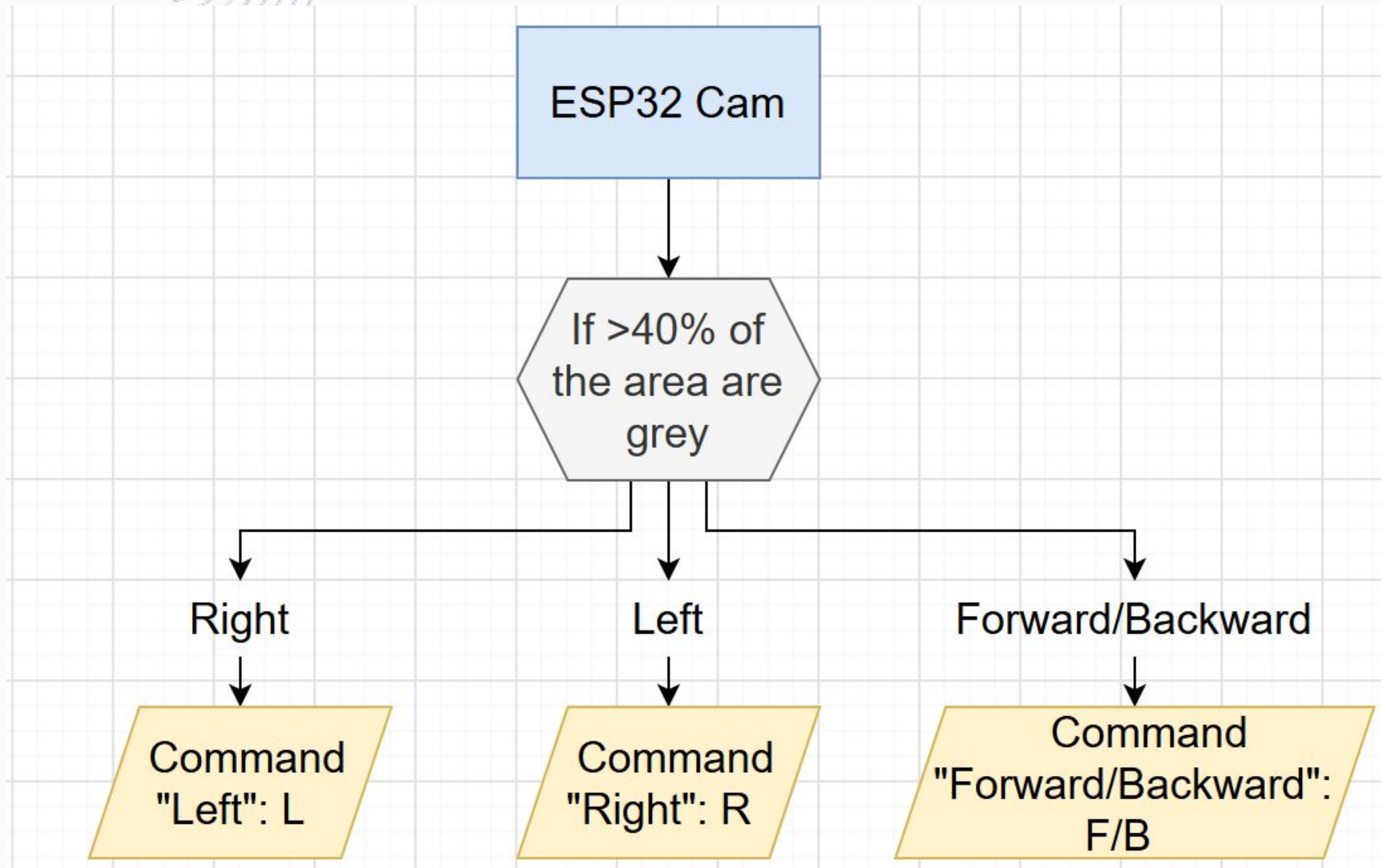
Examples:

128, 128, 128 -> Grey

120, 122, 119 -> Grey

40, 120, 40 -> Non-grey

Ground Egress Motion



Camera Colour Classification Testing

CanSat ESP32-CAM

RGB Camera Image

Reload this page to capture a new image.



Grey / Non-Grey Mask

Reload the page to capture a new image.



Camera Colour Classification Logic

CanSat ESP32-CAM

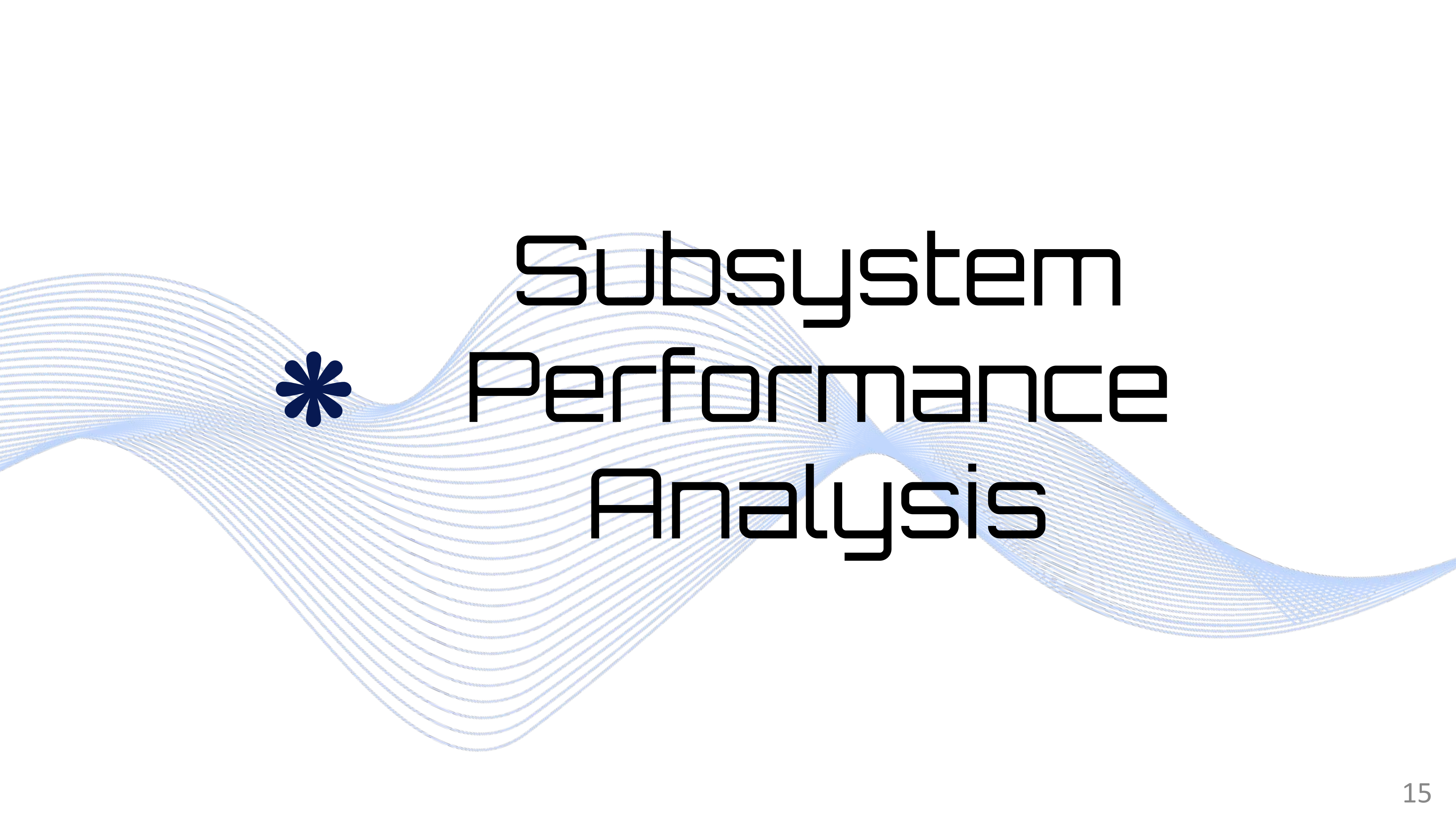
RGB Camera Image

Reload this page to capture a new image.



```
const int GREY_THRESHOLD = 40;
```





Subsystem * Performance Analysis

Electrical circuit logic

- **Circuit Architecture: Design Approach**

- **Arduino-centric integration, built module by module**

The circuit was designed around the **Arduino Nano** as the single point of connection for every subsystem. All sensor data, camera communication, and actuator control pass through it, rather than subsystems talking to each other directly. This kept the wiring logic simple

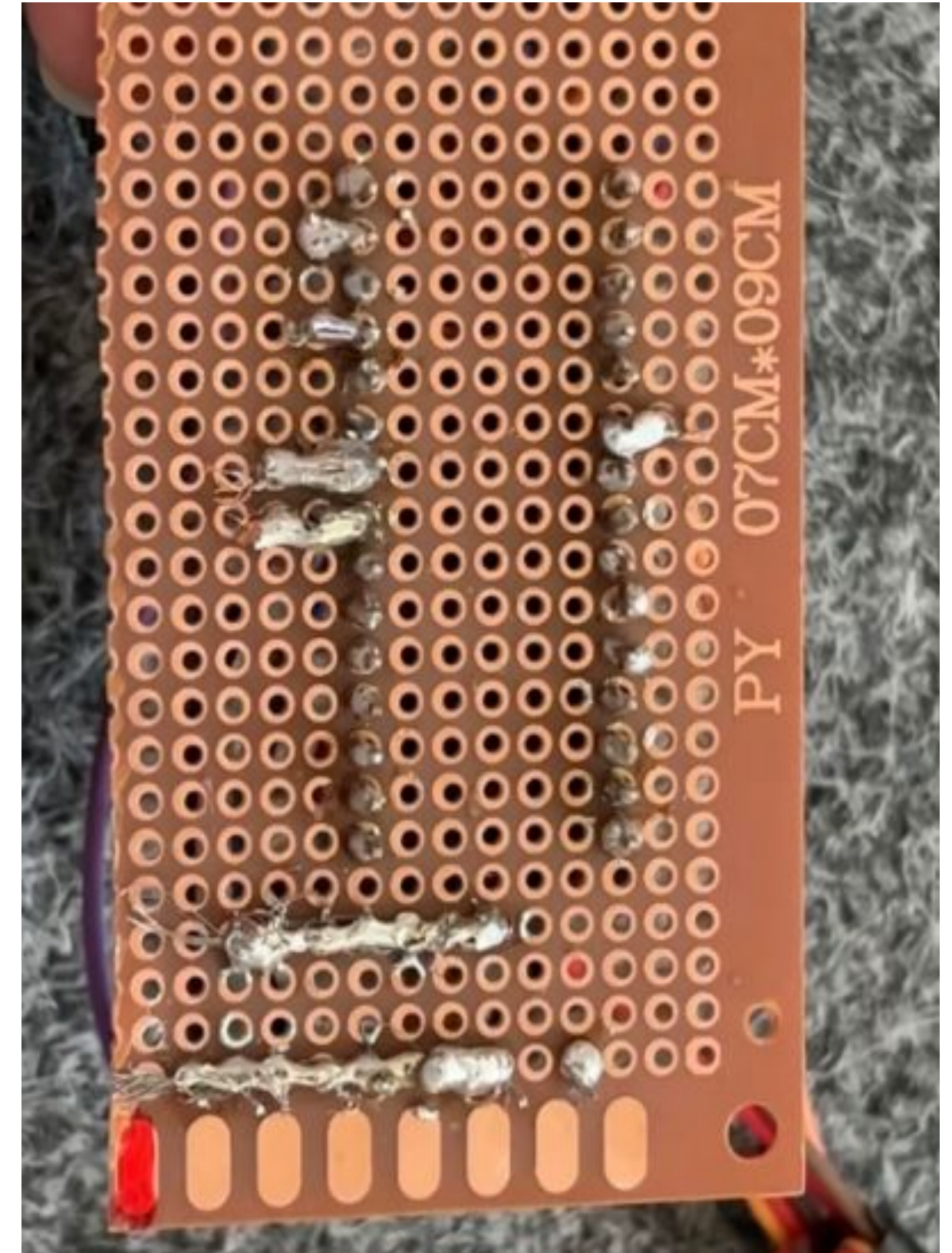
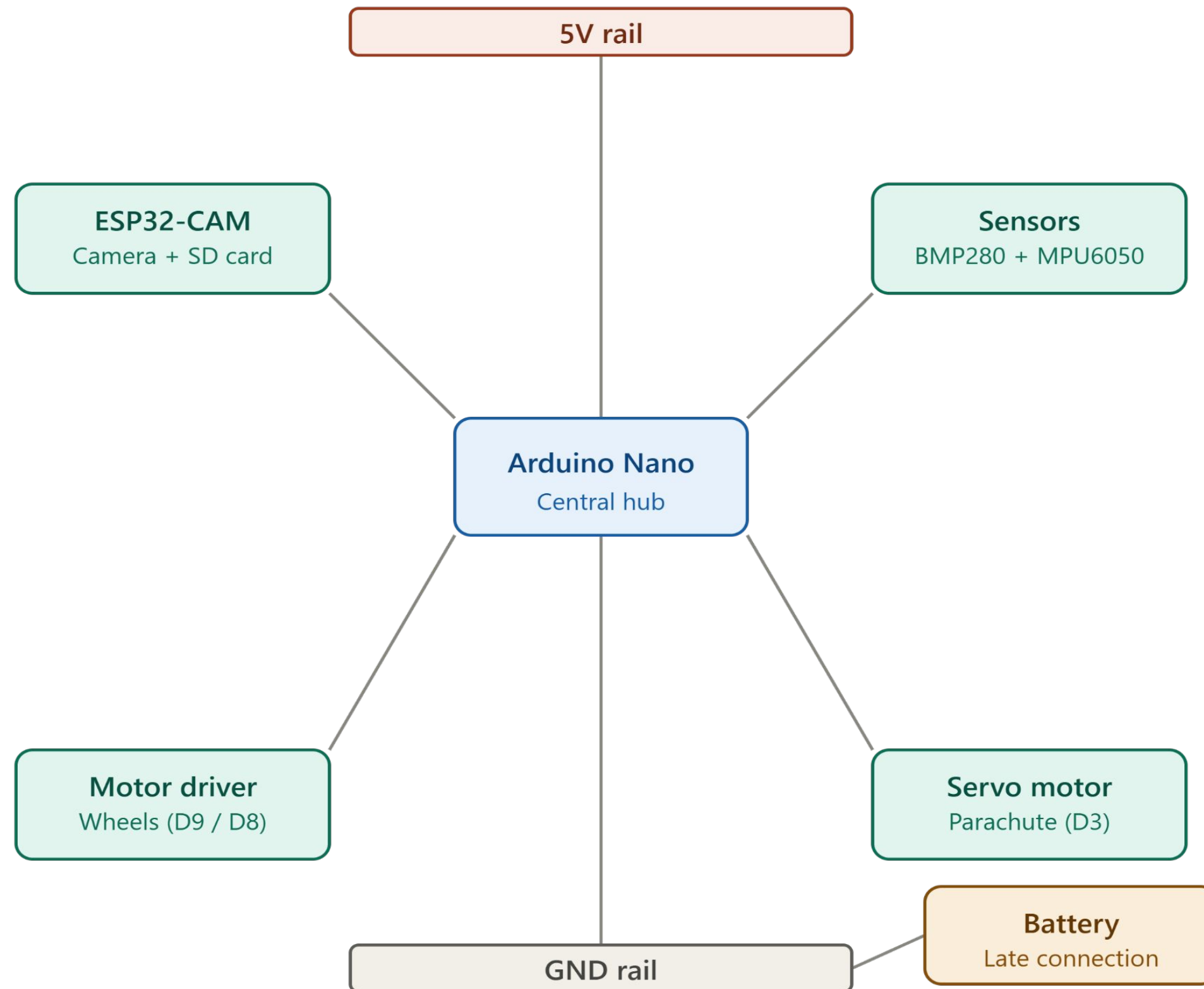
- **Power Distribution & Soldering Methodology**

- **A shared two-rail bus instead of point-to-point wiring**

Rather than running an individual power and ground wire from the Arduino to every component, the perfboard was built around two continuous copper rails: one carrying 5V, one carrying GND.

A single wire from the Arduino fed each rail, and every subsystem tapped into the same rail line

Electrical Circuit Logic



Subsystem Performance Analysis — Electrical

- **Parallel power distribution:** performed as designed, no faults
- **BMP280 + MPU6050 on shared I2C bus:** fully wired and functional
- **ESP32-CAM:** fully wired (incl. RX-line protection resistor), functional through launch
- **Motor driver:** one unit failed pre-launch (component damage), driver mounted directly on perfboard as the size-constraint fallback
- **Parachute-release servo:** not integrated; root cause was a torn signal wire during final handling, not a design or soldering flaw
- **Wiring architecture** (twin copper GND/5V rails + twisted-wire splices) held up structurally, but confirmed that very fine gauge wire (needed for the tiny component pins) is fragile under repeated handling

Recovery Outcome

CanSat Body	Intact, no structural damage from landing
Electrical system	Fully intact post-landing. Every wire, joint, and mounted component found in the same state as pre-launch.
Sensors (BMP280, MPU6050) and Camera (ESP32-CAM)	Intact and connected
Motors	motors mechanically intact; the one non-functional driver channel was a pre-existing condition, not landing-induced
Wheels	detached from the body due to the impact of the fall





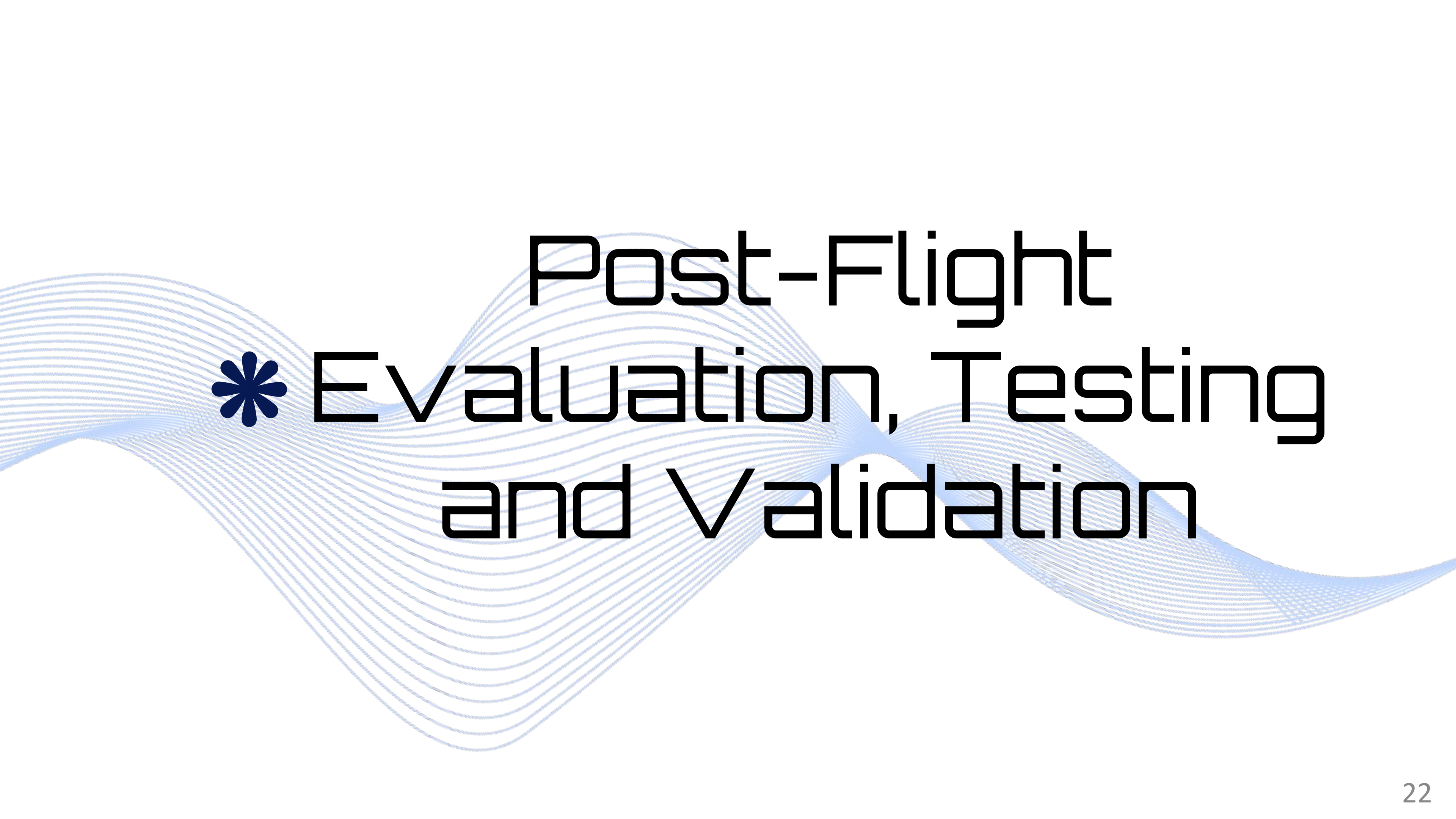
Flight Data Analysis

Firmware Integration & Testing

Modules	Purpose	Completion
Nano + MPU + BMP	Temp, Pa, altitude measurement	-
Nano + Sensors + SD Card	Data Logging (CSV) for velocity and altitude analysis	-
ESP32 Cam + SD Card	Colour Classification + Image Storage	-
Nano + Motor Driver	Motor control	-
Nano + Servo Motor	Parachute Detachment	-
Nano + ESP32	Communication connection	

Parameters	Values
Height	100 m
Duration	16 s
Pressure (100m)	933.44 hPa

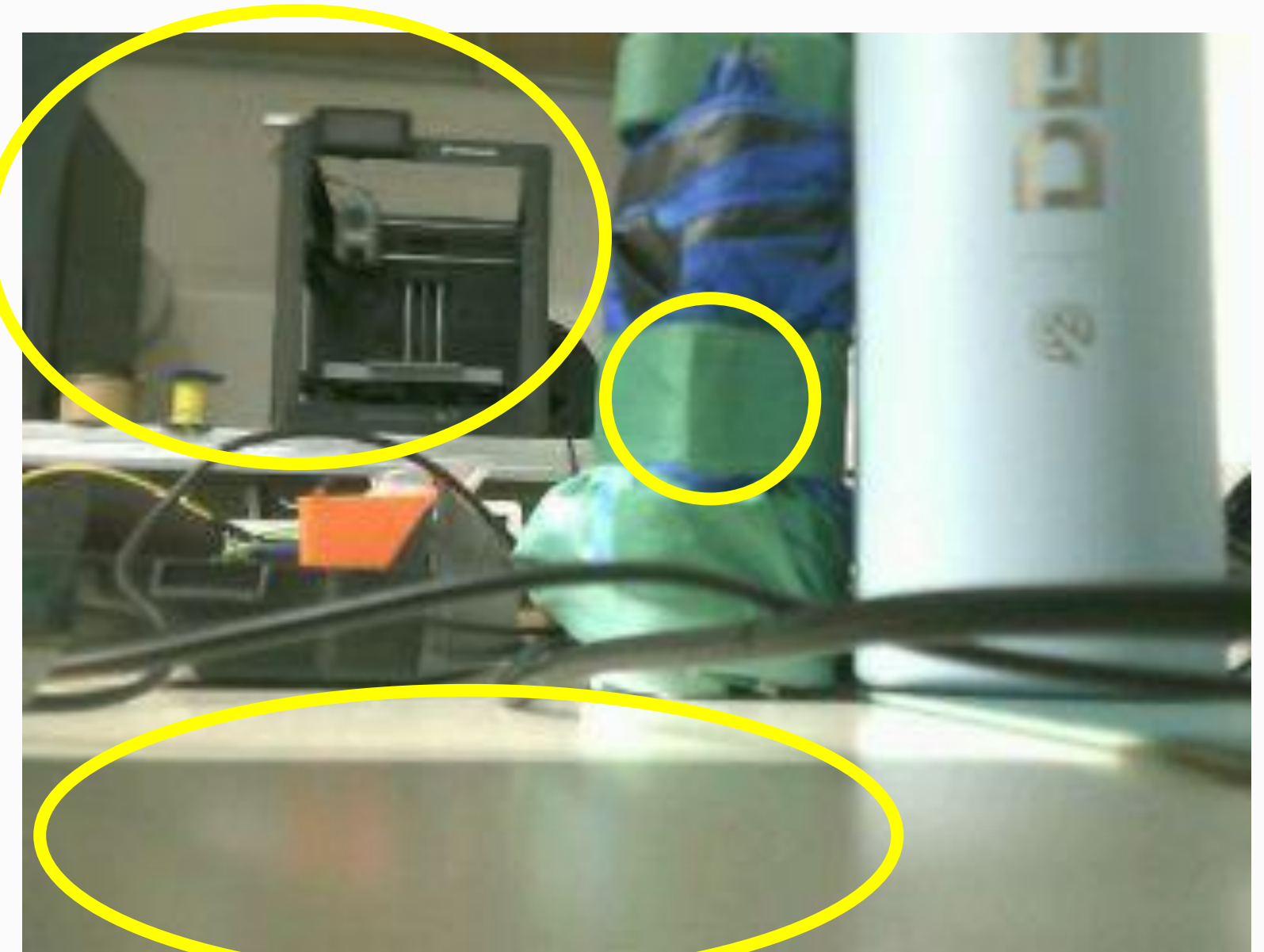
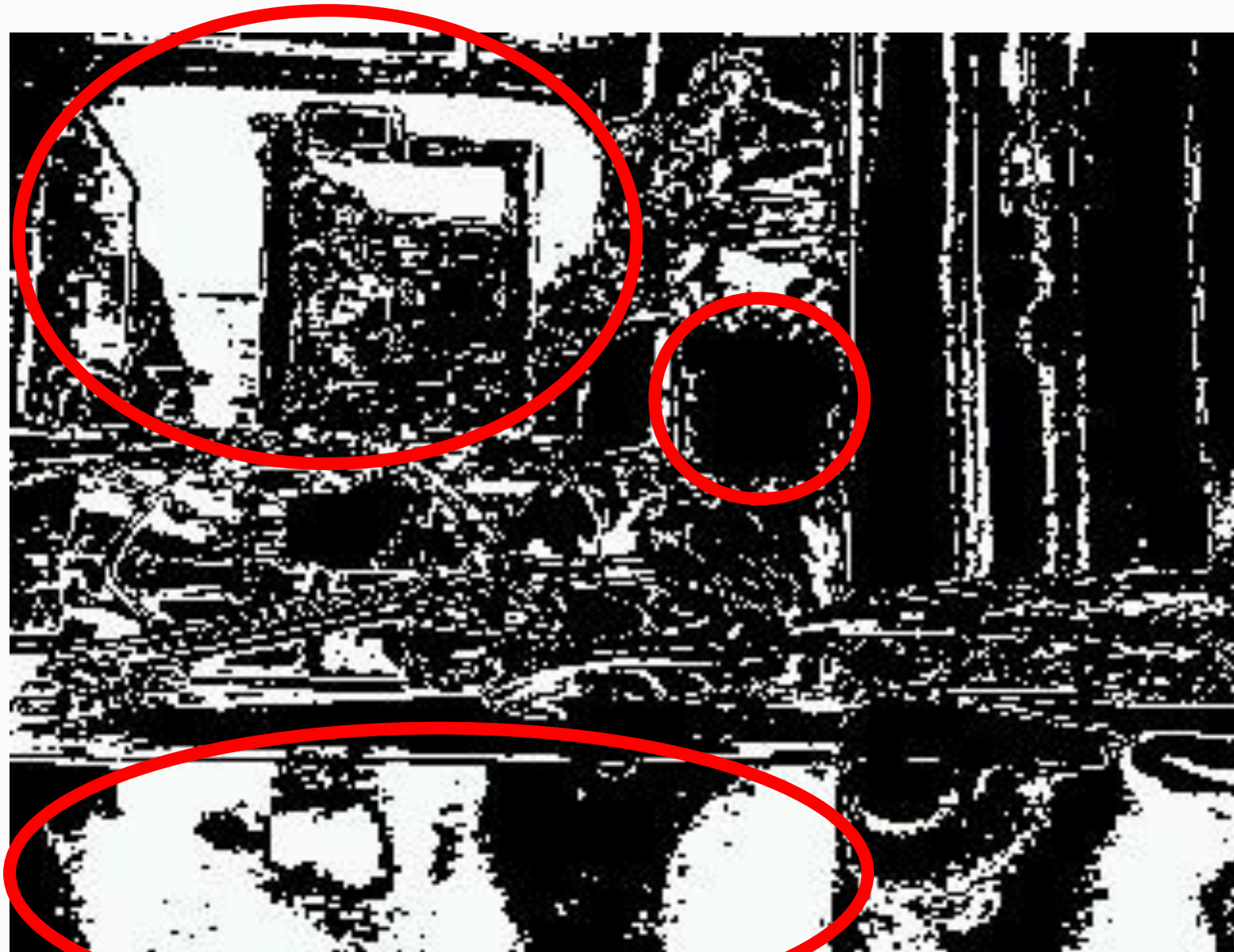
(Not enough time for preparation)



Post-Flight * Evaluation, Testing and Validation

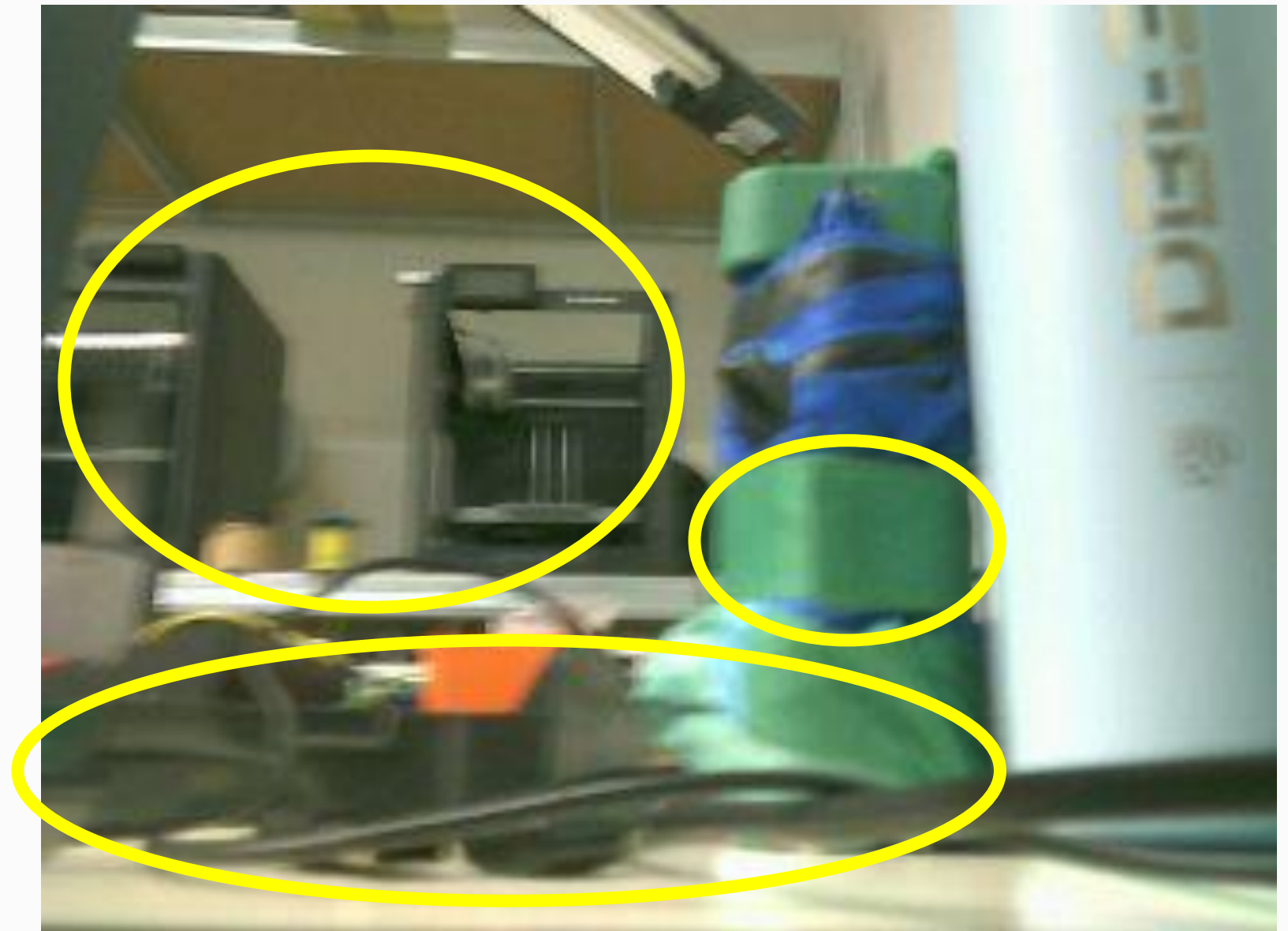
Object Detection & Avoidance - Step 1

- **Version 1:** Green & Non-green area classification
- Diagnosis: The camera could not distinguish between actual green and non-green objects
 - Color inversion trap: the esp32 and OV2640 read data in two different formats which swapped the RGB values; Auto White Balance (AWB) shifted colour code (e.g., green/yellow to white)



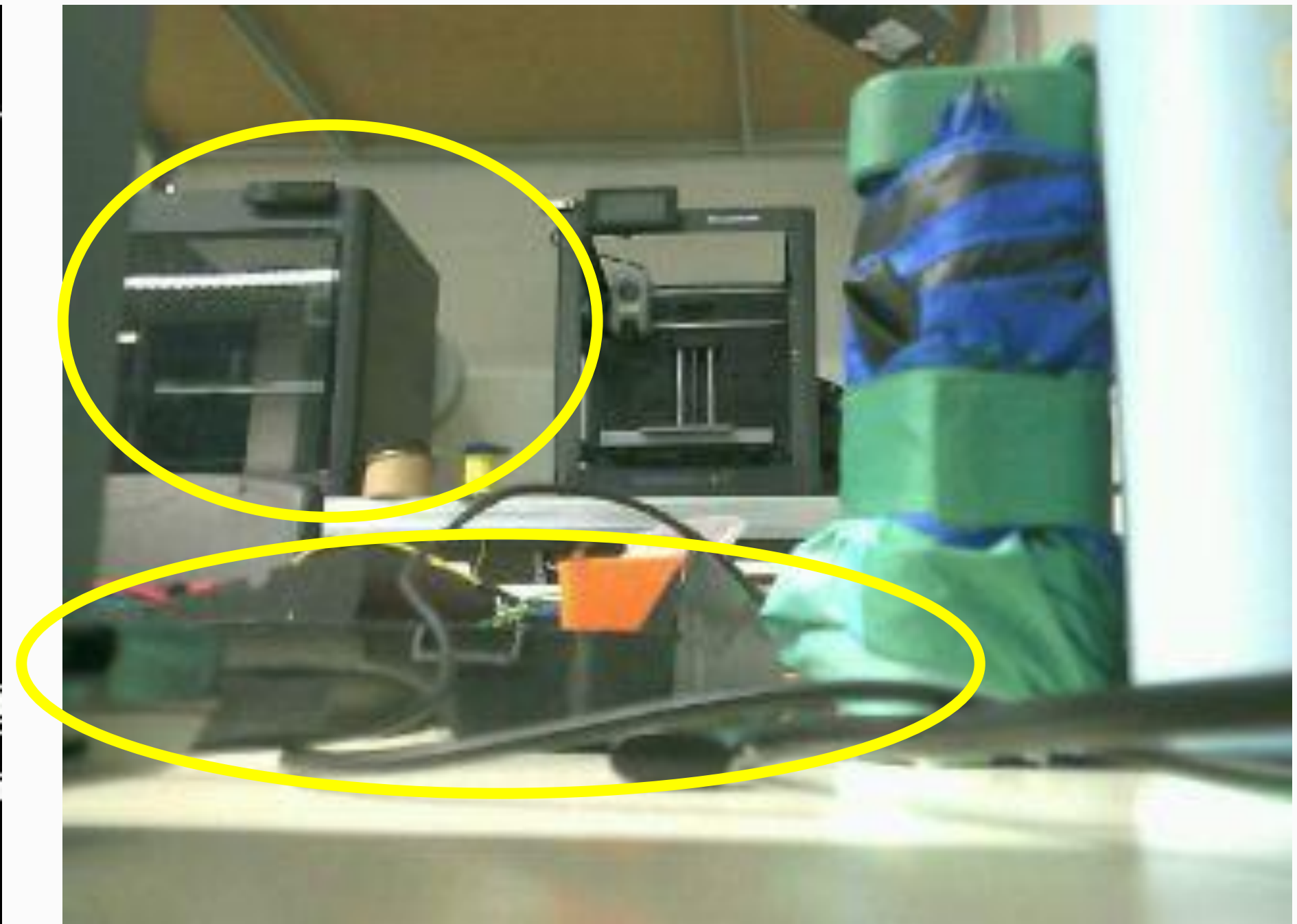
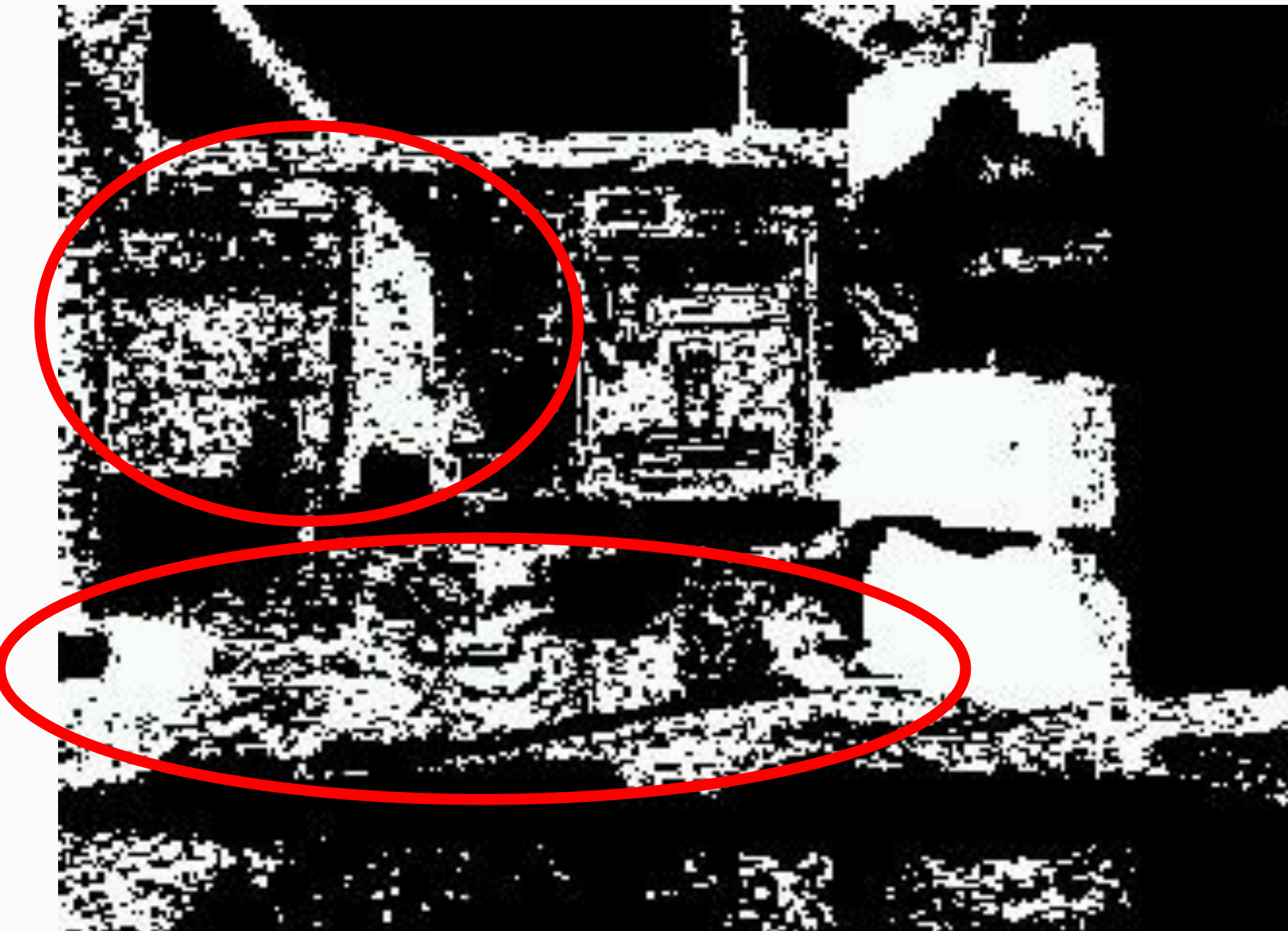
Object Detection & Avoidance - Testing & Validation

- Version 2: Solved colour inversion issue
- Diagnosis: Dark and black colour are misidentified as green objects
 - Green colour RGB value (565) ~ black colour with sensor noise (e.g., 373), the disturbed black colour code bypassed the threshold.



Object Detection & Avoidance - Testing & Validation

- Version 3: Improved precision of colour classification



Checking: Real-time Image Storage in SD Card

sketch_aug13a | Arduino IDE 2.3.10

File Edit Sketch Tools Help

sketch_aug13a.ino

1 #include <Arduino.h>

Output Serial Monitor X

Not connected. Select a board and a port to connect automatically.

ESP32-CAM SD STORAGE TEST

INITIALISING CAMERA

PSRAM detected.
Camera initialized successfully.

INITIALISING SD CARD

SD Card mounted successfully.

Starting capture loop in 2 seconds...

Writing frame (3920 bytes) to /DCIM/IMG_00001.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00001.jpg (3920 bytes)

Writing frame (4243 bytes) to /DCIM/IMG_00002.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00002.jpg (4243 bytes)

Writing frame (4316 bytes) to /DCIM/IMG_00003.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00003.jpg (4316 bytes)

Writing frame (4513 bytes) to /DCIM/IMG_00004.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00004.jpg (4513 bytes)

Writing frame (4500 bytes) to /DCIM/IMG_00005.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00005.jpg (4500 bytes)

Writing frame (4541 bytes) to /DCIM/IMG_00006.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00006.jpg (4541 bytes)

Writing frame (3141 bytes) to /DCIM/IMG_00007.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00007.jpg (3141 bytes)

Writing frame (5581 bytes) to /DCIM/IMG_00008.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00008.jpg (5581 bytes)

Writing frame (3797 bytes) to /DCIM/IMG_00009.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00009.jpg (3797 bytes)

Writing frame (2894 bytes) to /DCIM/IMG_00010.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00010.jpg (2894 bytes)

Writing frame (3221 bytes) to /DCIM/IMG_00011.jpg ...
SUCCESS: Saved and Flushed /DCIM/IMG_00011.jpg (3221 bytes)

Writing frame (6001 bytes) to /DCIM/IMG_00012.jpg ...

DCIM

USB Drive (D:) > DCIM

Search D

New

Name Date modified Type Size

IMG_00001 JPG File 4 KB

IMG_00002 JPG File 5 KB

IMG_00003 JPG File 5 KB

IMG_00004 JPG File 5 KB

IMG_00005 JPG File 5 KB

IMG_00006 JPG File 5 KB

IMG_00007 JPG File 4 KB

IMG_00008 JPG File 6 KB

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IMG_00010 JPG File 3 KB

IMG_00011 JPG File 4 KB

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IMG_00015 JPG File 4 KB

AutoSave Off

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File Home Insert Draw Page Layout Formulas Data Review

Paste

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B I U

Clipboard Font

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the cc

A1

fx Image_ID

	A	B	C	D	E	F	G
1	Image_ID	Timestamp	Green_Pixe	Total_Pixel	Green_Percentage		
2	00001	3717	11334	76800	14.76		
3	00002	9210	10579	76800	13.77		
4	00003	14252	11169	76800	14.54		
5	00004	19205	11291	76800	14.7		
6	00005	24247	10446	76800	13.6		
7	00006	29290	14256	76800	18.56		
8	00008	39285	18560	76800	24.17		
9	00009	44237	10589	76800	13.79		
10	00010	49280	21017	76800	27.37		
11	00011	54322	16320	76800	21.25		
12	00012	59275	19493	76800	25.38		
13	00013	64317	15663	76800	20.39		
14	00014	69270	17923	76800	23.34		
15	00015	74312	18255	76800	23.77		
16	00016	79355	16431	76800	21.39		
17	00001	3807	23460	76800	30.55		
18	00002	9390	22124	76800	28.81		
19	00003	14342	22820	76800	29.71		
20	00004	19385	21666	76800	28.21		
21	00005	24337	21807	76800	28.39		
22	00006	29380	22011	76800	28.66		
23	00007	34332	21320	76800	27.76		
24	00008	39375	20576	76800	26.79		
25	00009	44417	11102	76800	14.46		
26	00010	49370	7987	76800	10.4		
27	00011	54412	11325	76800	14.75		
28	00012	59365	13020	76800	16.95		
29	00013	64407	12260	76800	15.96		
30	00014	69360	12583	76800	16.38		
31	00015	74402	12956	76800	16.87		
32	00016	79355	12536	76800	16.32		
33	00017	84397	12236	76800	15.93		

Step 2: Integration of Objective Detection & Navigation

Logic:

1. Identify the lower region of the image (bottom 60%) that contains 80-100% white pixels to be grassland. If yes: Movement: F (Foward) Else: Movement: S (Stop)
2. Identify the concentrated black pixel region in the image. If on the left: Movement: R (right)

If on the right: Movement: R (right) If at the center, compare left vs right pixels concentration, e.g.,

55% right, 45% left: Movement: L (left)

Step 2: Testing & Validation

Sample Display:

FRAME # 00038 | Lower ROI White Coverage: 92.7%

ENVIRONMENT: Valid Grassland terrain confirmed.

COMMAND SENT : >>> [F] <<<



(Green Mat) Demonstration

```
AI Thinker ESP32-CAM
motioncommand.ino
Output Serial Monitor X
Message (Enter to send message to 'AI Thinker ESP32-CAM' on 'COM5')

-----
FRAME #00036 | Lower ROI White Coverage: 57.8%
ENVIRONMENT : Non-Grassland detected (<80% green).
COMMAND SENT : >>> [ S ] <<<
-----

FRAME #00037 | Lower ROI White Coverage: 84.1%
ENVIRONMENT : Valid Grassland terrain confirmed.
COMMAND SENT : >>> [ R ] <<<
-----

FRAME #00038 | Lower ROI White Coverage: 92.7%
ENVIRONMENT : Valid Grassland terrain confirmed.
COMMAND SENT : >>> [ F ] <<<
-----

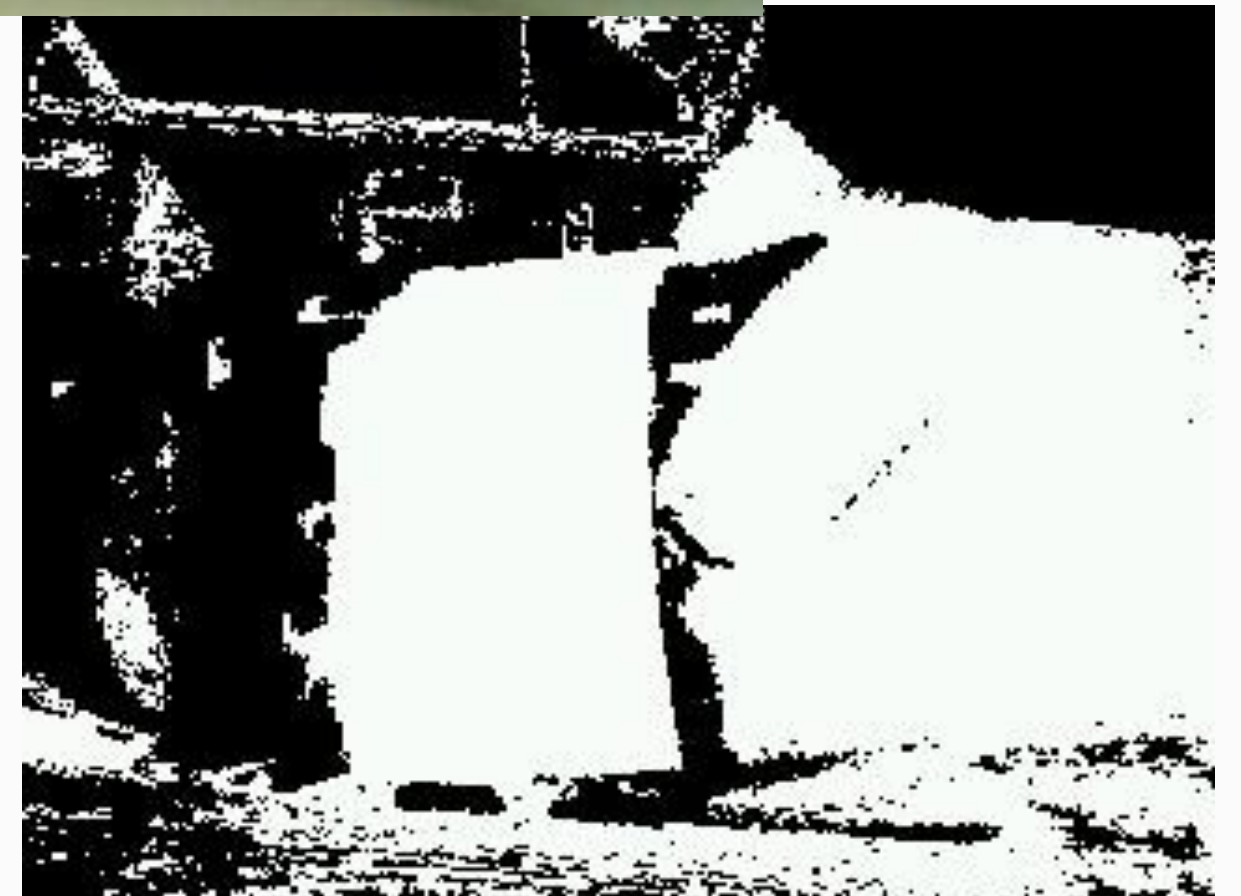
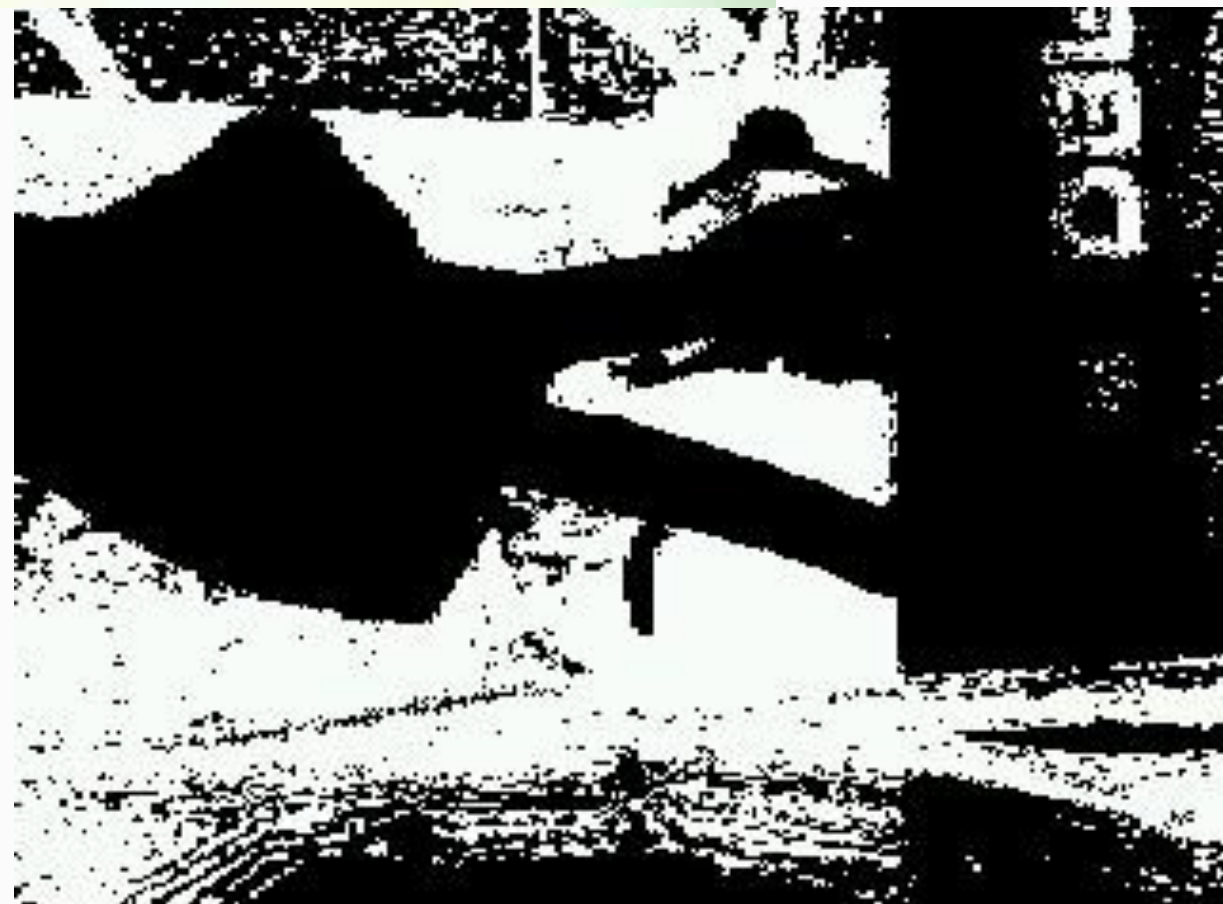
FRAME #00039 | Lower ROI White Coverage: 54.3%
ENVIRONMENT : Non-Grassland detected (<80% green).
COMMAND SENT : >>> [ S ] <<<
-----

FRAME #00040 | Lower ROI White Coverage: 54.1%
ENVIRONMENT : Non-Grassland detected (<80% green).
COMMAND SENT : >>> [ S ] <<<
-----

FRAME #00041 | Lower ROI White Coverage: 59.0%
ENVIRONMENT : Non-Grassland detected (<80% green).
COMMAND SENT : >>> [ S ] <<<
-----
```

Step 2: Testing & Validation

Sample Display: [S] (Stop)



Sensors Data Acquisition & Logging

Purpose:

1. CANSAT altitude record
2. Condition Monitoring
3. Data analysis in Post-Flight Evaluation

- Good for understanding the aerodynamics of cansat, surrounding environment, and CANSAT Maintenance
- Format: Easy to be stored in a txt file

```
8.3,28.7,1020.34,-58.89,0.24
4263,-0.16,0.97,-0.01,-1.04,1.02,0.15,28.2,28.7,1020.34,-58.87,0.26
4363,-0.16,0.97,0.00,-1.16,0.92,0.23,28.3,28.7,1020.34,-58.90,0.23
4463,-0.20,0.95,-0.07,7.92,-0.09,0.17,28.2,28.7,1020.34,-58.90,0.23
# Calibrating Ground Pressure... DONE! Ground Baseline: 1020.31 hPa
Timestamp_ms,AccX_g,AccY_g,AccZ_g,GyroX_dps,GyroY_dps,GyroZ_dps,MPU_Temp_C,BMP_Temp_C,Press_hPa,AbsAlt_m,RelAlt_m
1163,-0.16,0.97,-0.01,-1.10,1.01,0.17,28.4,28.8,1020.32,-58.68,-0.06
1262,-0.15,0.97,-0.02,-1.22,0.84,0.34,28.4,28.8,1020.32,-58.71,-0.08
1362,-0.16,0.96,-0.01,-1.04,1.04,0.32,28.5,28.7,1020.32,-58.71,-0.08
```

Altitude Control

International Barometric Formula

$$h = 44330 \times \left(1 - \left(\frac{P}{P_0} \right)^{\frac{1}{5.255}} \right)$$

- Calculation of relative pressure referenced from the sea-level pressure (1013.25 hPa)
- Built-in library function: `readAltitude()` - estimates

Sensors Data Acquisition & Logging

```
0.15,-0.08
2577,-0.15,0.97,0.16,-1.10,0.99,0.21,27.9,27.5,1019.29,-50.15,-0.08
2677,-0.15,0.97,0.15,-1.24,1.01,0.31,27.9,29.29,-50.15,-0.08
2677,-0.15,0.97,0.15,-1.24,1.01,0.31,27.9,27.5,1019.29,-50.15,-0.08
Nano operational. Streaming telemetry to ESP32...
378,-0.15,0.96,0.16,-1.21,0.92,0.35,27.9,27.4,1019.28,-50.08,0.01
478,-0.15,0.96,0.16,-1.21,1.05,0.26,27.8,27.4,1019.28,-50.08,0.01
577,-0.15,0.96,0.16,-1.19,0.93,0.24,27.8,27.4,1019.28,-50.08,0.01
677,-0.15,0.97,0.15,-1.27,1.08,0.40,27.9,27.4,1019.28,-50.08,0.01
777,-0.16,0.98,0.16,-1.21,1.04,0.38,27.9,27.4,1019.28,-50.04,0.04
877,-0.15,0.97,0.16,-1.28,0.89,0.21,27.9,27.4,1019.28,-50.04,0.04
977,-0.15,0.96,0.16,-1.11,0.99,0.37,27.9,27.4,1019.28,-50.04,0.04
1077,-0.15,0.98,0.16,-1.02,0.90,0.24,27.8,27.4,1019.28,-50.04,0.04
1177,-0.15,0.97,0.16,-1.11,0.98,0.32,27.9,27.4,1019.28,-50.04,0.04
1277,-0.15,0.96,0.16,-1.16,0.98,0.37,27.9,27.4,1019.27,-50.02,0.08
1377,-0.15,0.97,0.15,-1.11,1.02,0.26,27.8,27.4,1019.27,-50.02,0.08
1477,-0.15,0.97,0.16,-1.04,1.04,0.23,27.9,27.4,1019.27,-50.02,0.08
1577,-0.15,0.97,0.15,-1.07,0.95,0.27,27.8,27.4,1019.27,-50.02,0.08
1677,-0.16,0.96,0.16,-0.93,0.96,0.26,27.8,27.4,1019.27,-50.00,0.10
1777,-0.15,0.97,0.15,-1.10,1.04,0.26,27.8,27.4,1019.27,-50.00,0.10
1878,-0.15,0.96,0.15,-1.10,0.92,0.24,27.9,27.4,1019.27,-50.00,0.10
1977,-0.15,0.96,0.16,-1.13,0.95,0.37,27.9,27.4,1019.27,-50.00,0.10
2077,-0.14,0.98,0.16,-0.98,1.01,0.29,27.9,27.4,1019.27,-50.00,0.10
2178,-0.15,0.97,0.16,-0.98,0.99,0.27,27.9,27.4,1019.27,-49.97,0.12
2277,-0.15,0.97,0.16,-1.13,1.15,0.24,27.9,27.4,1019.27,-49.97,0.12
2377,-0.15,0.96,0.15,-0.98,0.79,0.40,27.8,27.4,1019.27,-49.97,0.12
2478,-0.15,0.97,0.16,-1.04,1.08,0.21,27.9,27.4,1019.27,-49.97,0.12
2577,-0.15,0.97,0.16,-1.13,0.98,0.34,27.9,27.4,1019.27,-49.97,0.12
2677,-0.15,0.97,0.16,-1.04,0.78,0.15,28.0,27.4,1019.26,-49.94,0.15
2778,-0.14,0.97,0.16,-1.16,0.95,0.24,27.9,27.4,1019.26,-49.94,0.15
2877,-0.15,0.97,0.16,-1.07,1.08,0.41,27.8,27.4,1019.26,-49.94,0.15
2977,-0.15,0.97,0.16,-1.22,1.07,0.34,27.9,27.4,1019.26,-49.94,0.15
3078,-0.15,0.97,0.15,-0.99,1.01,0.31,27.9,27.4,1019.26,-49.94,0.15
3177,-0.15,0.97,0.16,-0.99,0.92,0.26,27.8,27.4,1019.26,-49.94,0.15
3277,-0.14,0.97,0.16,-1.18,1.05,0.34,27.9,27.4,1019.26,-49.94,0.15
3378,-0.16,0.97,0.16,-1.04,1.05,0.23,27.9,27.4,1019.26,-49.94,0.15
3477,-0.15,0.97,0.15,-1.16,1.02,0.29,27.9,27.4,1019.26,-49.94,0.15
3577,-0.14,0.96,0.16,-1.22,0.95,0.29,27.9,27.4,1019.26,-49.92,0.17
3678,-0.16,0.97,0.15,-0.98,0.98,0.31,27.8,27.4,1019.26,-49.92,0.17
```



Real-time wired transmission of data from sensors



Failed to Store the data to the SD Card in ESP32 Camera Module

Cause of failure:

1. Loose connection in hardware integrations
2. Power Supply Brownouts (huge current spike)
3. Problem of Readability of SD Card Format

Lessons Learned - *Technical*

- Understand material limitations/properties before committing to a design (inflexible PLA).
- Be careful when testing sensors, motors, and motor drivers.
- Tape is a practical solution, but not trustable

- AI-Assisted Development & Architectural Refactoring
- Agile development & rapid-prototyping
- Cross-module integration
- Memory optimization
- Core EPS architecture (parallel power distribution, shared I2C bus) withstood real flight and impact conditions with zero faults

Lessons Learned - *Project Management*

- Communicate early about each other's experience and expertise.
- Include other team members in your progress, regardless if it's ahead or behind expected schedule.

Recommendations for future missions

- Provide additional strategies/ more flexibility/encouragement to tackle primary goals when crucial modules lost their functionalities
- Start soldering earlier
- Keep a launch-day contingency kit (spare wire, iron, multimeter) ready on-site
- Hardware Supply (5 Xbee, 4 explorer boards, 4 BMP sensors tested)



Thank you

CREDITS: Slide template and graphics were created by **CANVA**

